A Modular, FHIR-Integrated Framework for LLM-Assisted Pre-Consultation

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Abstract

We describe a modular reference implementation for LLM-assisted pre-consultation with deterministic questionnaire control and structured FHIR write-back. Synthetic verification covers interoperability, end-to-end operation, module replacement, and local concurrency. Questionnaire definitions are exported from versioned source JSON; model adapters support commercial and local inference; write-back retains patient, EHR, model, and clinician provenance. Explicit run selection and source hashes generate the reported tables and numerical values. The selected provisional end-to-end evidence contains 150 sessions with completion of 100.0% (150/150). Written-resource checking reports 350 verification errors, including 350 resources left unverified after validator exceptions. Scenario

review and independent audits remain pending. These findings describe technical verification and its limitations, not diagnostic accuracy, clinical safety, or national deployment readiness.

Keywords: FHIR, Pre-consultation, Conversational AI, Clinical Documentation

1 Introduction

Low-acuity visits place a growing burden on unscheduled care worldwide.Pines et al. (2011) In Taiwan, despite low-cost access under National Health Insurance, its emergency departments remain chronically crowded, and about one-third of visits are non-emergent.Huang et al. (2019) This crowding prolongs waits, worsens outcomes, and drives clinician burnout.Morley et al. (2018); Ghaseminejad et al. (2025)

Large language models (LLMs) now support numerous clinical tasks, including ambient scribing, EHR documentation, and clinical decision support.Nair et al. (2026) Agentic systems extend these capabilities through planning, tool use, and multi-step action, enabling closer integration with health-system workflows and conversational diagnosis.Tu et al. (2025) However, although various benchmarks now evaluate LLMs on realistic tasks and interactive encounters, most agentic systems remain confined to simulated settings.Liu et al. (2026)

Here, we present XXX, an LLM-based agent for adults presenting in Taiwanese urgent care centers. XXX conducts structured history-taking by text or voice, drafts a clinician-ready note, and supports decision-making with a ranked differential and not-to-miss conditions, alongside suggested examinations and imaging. To limit hallucination, the LLM only extracts clinical facts linked to the patient's own words, while safety screening, question selection, and diagnostic ranking follow auditable rules. XXX connects to EHRs via SMART on FHIR and was evaluated on benchmark.

2 Methods

2.1 Scope and implemented architecture

We evaluated a reference implementation for structured pre-consultation using synthetic patients. The objective was software verification: interchange of clinical data, completion of the interview-to-write-back workflow, replacement of model adapters, and concurrent operation. Diagnostic accuracy, triage accuracy, clinical note quality, and comparisons with clinicians were outside scope.

The implementation comprises a Vue patient interface, a clinician interface, a FastAPI backend, a local retrieval component, and a FHIR server (Fig. 1). The patient interface uses an invitation exchanged for a scoped session cookie. The clinician launcher binds the patient and visit before the interview. SMART launch is a separate integration surface; it is not equivalent to exchanging the patient invitation.

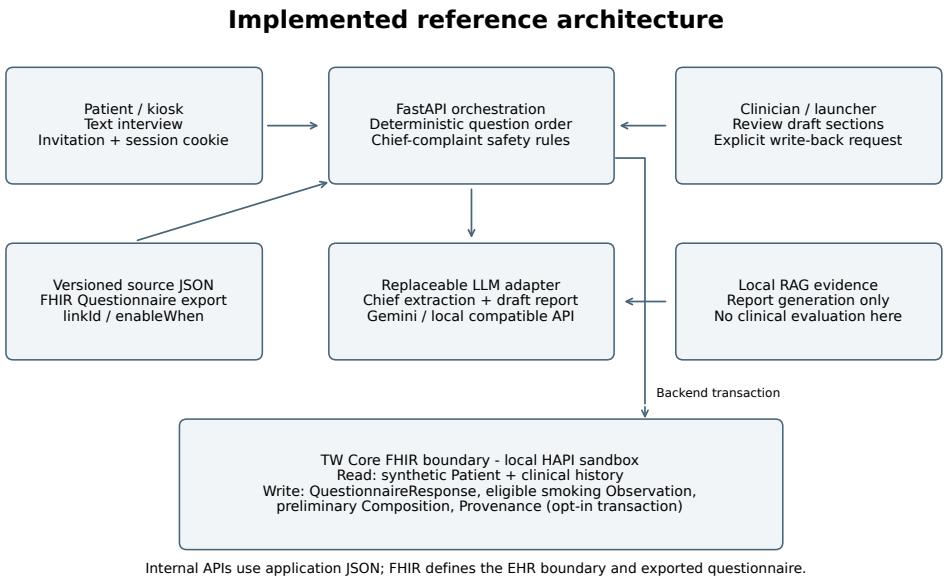


Fig. 1 Implemented modules and data boundaries. The running questionnaire engine consumes versioned application JSON; an encoder exposes a corresponding FHIR Questionnaire. Internal application APIs are JSON contracts, whereas the EHR boundary uses FHIR. Arrows do not imply a universal FHIR-native internal bus.

Experiment instances use an isolated SQLite database and an explicit loopback development authorization bypass. Consequently, these runs do not establish production authorization or institutional security assurance.

Question selection is deterministic. The engine asks the next enabled question from the configured source and skips fields already populated from the EHR. It does not delegate every conversational turn to an LLM. At entry, deterministic rules and evidence-checked model extraction screen the chief complaint. Failure of semantic screening leads to manual handoff. In this questionnaire engine, later answers are not repeatedly screened; this is a limitation requiring separate clinical review. At routine completion, retrieval supplies report evidence and the LLM drafts record sections and decision-support suggestions. These suggestions are not evaluated as diagnoses or published as orders.

2.2 Questionnaire and write-back contracts

The encoder maps source field names to stable `linkId` values, source options to coded answers, and representable conditional branches to `enableWhen`. FHIR Questionnaire separates the form definition from its answers in QuestionnaireResponse.HL7 International (2019) The exported canonical version includes a source-content digest. Extensions retain application-specific presentation, prefill, and semantic metadata; generic FHIR consumers must explicitly support these extensions to reproduce all

Table 1 Implemented write-back resources and assertion boundaries.

Resource	Content and boundary
QuestionnaireResponse	Ordered answers with patient-reported, EHR-prefilled, or unattributed source markers; canonical questionnaire reference.
Observation	Smoking status only when explicitly selected by the patient; unmapped or EHR-prefilled text is not promoted to a new coded patient statement.
Composition	Preliminary narrative with reviewed sections and references to the questionnaire response and eligible observation.
Provenance	Patient informant, AI assembly, and clinician review roles; experiment confirmation is simulated, not clinical approval.

application behavior. A separate evaluator derives expected interview paths from the exported definition and scripted answers, including prefilled fields. Equivalence is tested for the supported answer grammar, not assumed for arbitrary EHR free text.

Structured write-back is an opt-in backend capability. A clinician-authorized request with confirmed sections creates a transaction containing a QuestionnaireResponse, a preliminary Composition, Provenance, and an eligible smoking-status Observation (Table 1). Conditional creation and stable transaction references support retry idempotency. The legacy Composition-only mode remains the default. Patient and encounter references, institution scope, and the configured FHIR issuer are checked before writing.

The transaction does not create differential-diagnosis Conditions, ClinicalImpressions, ServiceRequests, or safety Flags. A coded smoking mapping and the provenance wording remain subject to terminology and clinical review. The questionnaire version is currently resolved at write-back rather than persisted at interview start, limiting historical reproducibility if source definitions change between those events. When application and FHIR conditional semantics disagree for stored answers, structured write-back fails closed rather than silently discarding a response.

2.3 Synthetic inputs and scenario review

Synthea generated a deterministic adult cohort.The MITRE Corporation (2026) The conversion pipeline localizes names and addresses to fictional Taiwanese values and assigns TW Core profiles to retained resources. Source seeds, reference date, tool checksums, mapping decisions, and dataset fingerprints are recorded with the data. Synthetic Taiwanese presentation does not make the underlying US-derived disease model representative of Taiwan. Terminology substitutions, dropped elements, and unresolved mappings are reported explicitly. Input validation is separate from validation of resources written by the system.

The scenario set contains 30 author-drafted scripts covering chest pain, headache, and abdominal pain, including intended safety stops and documented rule-coverage

gaps. The benchmark named in the initial plan was unavailable; scripts were instead authored for this study with provenance to earlier synthetic feasibility cases where applicable. An initial background-matching procedure missed contradictory histories, so each scenario received a separate source-level Synthea module. Generated patient output is not manually rewritten to force a match. Mechanical audits compare the authored active conditions, medications, allergies, and smoking history with the resulting resources. These are deliberately constructed software fixtures, not a population sample or a test of clinical plausibility. Source free text uses explicit local codes where no reviewed standard mapping exists. Empty lists and generator-assigned verification metadata do not constitute clinical absence or physician confirmation. Symptom plausibility and answer completeness still require human review.

Scripts remain provisional until their status is marked validated by a human and two independent physician decisions are recorded against the final content hashes. Patient bindings and expected paths are part of that hashed content. The reviewer packet presents source facts, answers, expected paths, and safety-rule expectations. At asset generation, 0 scripts have the required recorded approvals. No automated test or model response substitutes for those decisions.

2.4 Verification procedures and outcomes

E1 captures resources read and written by the live API workflow. The HL7 validator runs with the pinned TW Core package and generated questionnaire and extension definitions. Structural/profile validation with terminology disabled is reported separately from terminology-enabled checks. The prespecified target is no errors on written resources. Validator exceptions and missing outcomes count as failed verification, not conformance. Round-trip comparison ignores only specified server metadata and assigned identifiers, while preserving substantive content differences. A separate seeded templated cohort broadens input coverage.

E2 uses the patient-facing HTTP API from invitation exchange through dialogue, report creation, clinician-confirmation simulation, write-back, and read-back. The simulated patient is a local Qwen model distinct from the system models. Its prompt permits only facts in the scenario and prohibits disclosure of diagnoses; the opening complaint is sent verbatim. Constrained output is checked against the displayed options before composing the kiosk answer. Invalid answers are recorded, not replaced with the expected answer. Runs retain transcripts, timing, trace, evidence, and token metadata. Run-created write-backs are removed before reusing a patient so later repetitions cannot read earlier interview answers.

Completion requires reaching write-back with an explicit completed or summary-ready workflow status. Original harness summaries count API write-backs; the paper's reproducible reanalysis excludes manual handoff and unknown workflow statuses from successful completion. Timing distributions retain all API write-backs. Protocol adherence compares the ordered recorded path with the independent expected path; intended early safety stops and unavailable paths have explicit exclusion counts. Fact traceability uses all persisted chief-complaint facts as the denominator, including facts missing evidence, and checks exact or normalized spans against patient utterances. It does not cover every statement in the final narrative or establish medical correctness.

Safety execution is a software functional check of expected rule identifiers; coverage-gap scripts cannot pass merely because their expected-rule list is empty. No sensitivity or specificity is estimated.

E3 changes module configuration, reruns workflow and interoperability checks, and records source fingerprints and configuration diffs. The one-time implementation work needed to introduce adapters is reported separately from changes between configurations. The commercial alternate uses the same vendor, while the local alternate changes both provider interface and model. These comparisons establish only the tested boundaries; an API-level frontend entry test does not establish embedded-HIS usability or a complete OAuth browser journey.

E4 replays captured patient turns using an asynchronous HTTP load generator. Its mock-model mode uses fixed model latency, while a separate real-model mode tests small concurrency levels. Replays assigned across synthetic histories are load fixtures, not clinically consistent scenario cohorts. The implementation differs from the planned k6/Locust tool but keeps the declared concurrency sequence and early-stop criterion. Latency percentiles, throughput, request errors, session failures, and FHIR-write latency are retained. Results describe the tested local host and worker count, not national capacity. The national peak-demand estimate remains an author-supplied requirement. E5, testing a national exchange-hub environment, is blocked pending access; no government endpoint was contacted.

Results are descriptive; repeated sessions from a script are dependent and are not independent clinical observations. Tables show actual denominators and medians with interquartile ranges. Token-derived cost covers observed, priced API calls only: asynchronous summaries, late timeout completions, and calls outside bounded telemetry may be missing. It is not a complete economic estimate. An explicit run-selection file and source hashes generate every result table, figure, and numerical macro; earlier failures and interrupted runs remain in the run inventory. No diagnostic benchmark or credentialed patient dataset is used.

3 Results

Evidence status: Provisional; physician scenario review pending. The tables distinguish observed technical outputs from clinical validation. Empty cells marked "–" denote an unavailable or unexecuted measurement, never a zero error count.

3.1 Interoperability

The corrected main input cohort contains 1,000 bundles and 430,017 resources. Independent input checking reports 0 errors and 1,058,802 warnings. The corrected report combines unchanged bundle outcomes with independent revalidation of every changed bundle, using content hashes and complete coverage checks. Offline validation disables network terminology resolution but can still use cached terminology evidence. These input results are separate from system-output conformance.

The selected E1 evidence comprises 200 templated sessions and 150 scenario sessions. It covers 1,348 written resources, with 350 reported verification errors and 1,750

Table 2 E1 written-resource validation. Exceptions count as failed verification; input validation is excluded from these counts.

Cohort	Resource	n	With errors	Errors	Warnings
templated batch	Composition	200	0	0	200
templated batch	Observation	198	0	0	0
templated batch	Provenance	200	0	0	800
templated batch	QuestionnaireResponse	200	200	200	0
e2 sessions	Composition	150	0	0	150
e2 sessions	Observation	100	0	0	0
e2 sessions	Provenance	150	0	0	600
e2 sessions	QuestionnaireResponse	150	150	150	0

Table 3 E1 workflow and semantic round-trip checks. Read rates use all sessions; write rates use sessions reaching a write attempt. Read-back uses returned resource references, while round-trip uses all captured submissions, including submissions with missing references or read-back.

Cohort	Sessions	Read	Write	Read-back	Round-trip
templated batch	200	100.0% (200/200)	100.0% (200/200)	100.0% (798/798)	100.0% (798/798)
e2 sessions	150	100.0% (150/150)	100.0% (150/150)	100.0% (550/550)	100.0% (550/550)

warnings (Table 2). This includes 350 resources left unverified after validator exceptions; this count is not the number of distinct exceptions. Each unverified resource contributes a failed check to the error total rather than being counted as conformant. The zero-error target is not established when an exception prevents a resource from being checked. A minimal base-FHIR example reproduced the pinned validator's incompatible-answer-type exception when a legal open-choice string answer is compared with a coded conditional answer. Preserving the original EHR text avoids inventing a patient selection, but leaves this verification gap unresolved. Warnings and input-validation outcomes are retained separately in the generated supplementary tables and raw result files.

Table 3 separates workflow success from resource fidelity. Successful transfer does not establish profile conformance. Positive terminology resolution of every coding remains unavailable; absence of a reported coding issue is only a proxy and is not presented as a resolution rate.

3.2 Workflow feasibility and replacement

The selected E2 run requested 5 repetitions per script and contains 150 attempted sessions. Completion is 100.0% (150/150), exact path agreement among comparable paths is 100.0% (100/100), and traceable extracted facts are 100.0% (446/446) (Table 4). These provisional measurements cannot be presented as physician-validated findings. Unbound scenarios, early stops, missing traces, and failure stages remain visible in the run summaries rather than being silently removed from the planned cohort.

Table 4 E2 provisional synthetic feasibility. Percentages include their actual numerator/denominator; path comparison excludes explicitly classified early stops. Times include the workflow through write-back.

Cohort	n	Completion	Path exact	Traceable facts	Turns median [Q1,Q3]	Seconds median [Q1,Q3]
Chest pain	50	100.0% (50/50)	100.0% (35/35)	100.0% (141/141)	15.5 [1.0, 17.0]	33.0 [15.1, 37.1]
Headache	50	100.0% (50/50)	100.0% (30/30)	100.0% (150/150)	14.5 [1.0, 15.0]	33.4 [9.1, 37.0]
Abdominal pain	50	100.0% (50/50)	100.0% (35/35)	100.0% (155/155)	13.0 [1.0, 15.0]	32.8 [14.1, 37.9]
Overall	150	100.0% (150/150)	100.0% (100/100)	100.0% (446/446)	15.0 [1.0, 15.0]	33.1 [12.4, 37.2]

Table 5 Additional E2 provisional software checks and observed usage. Rule agreement is a functional test, not clinical sensitivity. Token and cost values cover the bounded recorded calls only; local hardware cost is excluded.

Overall provisional measurement	Observed value
Red-flag scripts: exact expected rules	83.3% (50/60)
Red-flag scripts: acceptable rules	83.3% (50/60)
Other scripts: no unexpected rules	100.0% (90/90)
Sessions with documented rule-coverage gaps	10
Sessions with recorded model usage	150
System input tokens/session, median	17,029
System output tokens/session, median	1,748
Simulated-patient tokens/session, median	17,522
Observed API cost/session, median USD	0.0175
Sessions contributing cost estimates	110
Invalid simulated-patient turns	50

Table 5 reports safety-rule agreement, documented coverage gaps, invalid simulated-patient turns, and observed usage. Random fact and transcript audit samples are exported with empty reviewer columns for independent review.

The selected E3 evidence includes 3 evaluated configuration rows (Table 6). An unavailable comparison is left unmeasured. Source fingerprints support the configuration-only comparison only when the code used by the compared runs is identical. Successful model invocation alone is insufficient evidence of interchangeability if completion or FHIR checks fail. Across these runs, 2 API write-backs still required manual handoff and are excluded from the completion numerator. Internal operation timeouts may allow record persistence without successful completion of every model or retrieval operation; token logs do not reliably represent those failures. The local system and simulated-patient models share GPU resources, so contention is included in the local configuration's latency.

3.3 Load and walkthrough evidence

Figure 2 reports the selected load-run evidence. Mock-model throughput does not estimate real-model capacity; the error threshold determines whether later planned

Table 6 E3 configuration comparisons, provisional. Code changes are measured after the one-time framework extension. API costs marked by an asterisk are partial recorded-call estimates, excluding hardware and omitted calls; their sample sizes include only sessions contributing a cost estimate. Reported errors include unverified resources after validator exceptions.

Configuration	Code lines	Completion	Path exact	Facts	Errors	Seconds	USD/session*
default	0	100.0% (30/30)	100.0% (20/20)	100.0% (89/89)	30	36.0	0.0184 (n=22)
a_gemini_alt	0	100.0% (30/30)	100.0% (20/20)	100.0% (119/119)	30	31.2	0.0097 (n=22)
b_ollama	0	93.3% (28/30)	90.9% (20/22)	100.0% (6/6)	30	202.2	0.0000 (n=20)

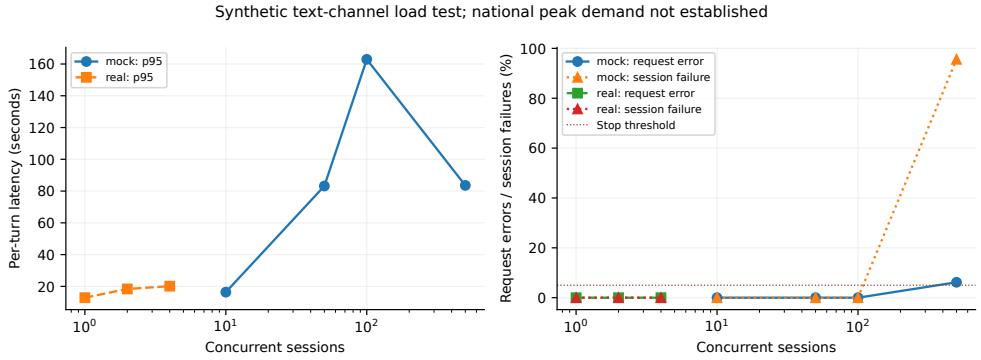


Fig. 2 Local synthetic load experiment. Missing modes or levels were not measured; the dotted line marks the declared early-stop threshold. Modes differ in replay workload and backend revision.

concurrency levels are attempted. At the last attempted concurrency of 500, 22 of 500 sessions completed. The request error rate was 6.18%, while the session failure rate was 95.6%. This mock run loaded the backend before the final conditional-write-back guard; it must not be represented as a performance measurement of the final backend revision. An interrupted E2 pilot overlapped this load run and used backgrounds later found inconsistent; it is preserved in the run inventory and excluded from the corrected cohort. The real-model run produced 7 API write-backs in 7 measured sessions at concurrency 1, 2, 4. This small replay workload uses a later backend revision; it is not a matched comparison with mock mode. No validated chest-pain script is currently available for the requested clinical walkthrough. The artifact directory therefore records that dependency instead of presenting an unreviewed example as clinician-approved.

4 Discussion

The implementation demonstrates a concrete separation between questionnaire control, model-assisted drafting, and EHR persistence. The principal contribution is an

inspectable contract and reproducible verification workflow. It is narrower than a universally FHIR-native conversational agent: questionnaire source JSON still controls the runtime, application-specific extensions carry additional semantics, and internal calls use application contracts. These boundaries should be explicit when adapting the framework to another institution.

Verification revealed issues that a successful end-to-end demonstration alone would miss. Free-text EHR values cannot safely be forced into coded questionnaire options. Conditional-semantic disagreement can discard information unless detected, and validator failures can prevent an interoperability conclusion even when a server accepts a transaction. Likewise, repeated synthetic sessions can contaminate later inputs through prior write-backs. Recording these failures and preserving their artifacts is necessary to distinguish repaired implementation defects from limitations of the evaluation environment.

The simulated-patient harness also required verification. A substring-based unknown-answer fallback could select a symptom option containing the same word as an uncertainty label. Affected runs were excluded and preserved before a full rerun with exact-option matching. This failure illustrates why API completion, path agreement, and traceable chief-complaint facts cannot establish fidelity of all subsequent simulated answers. Independent transcript review remains necessary.

The study cannot establish clinical safety, utility, or readiness for deployment. Scenarios are author-drafted and await physician review; textual evidence matching does not validate diagnostic interpretation; the simulated patient can still hallucinate; and synthetic demographics do not reproduce local disease burden. Only chief-complaint safety screening is active in the default workflow. Questionnaire terminology and smoking-status mappings require clinical governance. The local sandbox bypasses production authentication, and concurrency measurements do not test multi-institution isolation, recovery, national networking, or the national exchange hub. Model aliases may change upstream even when their reported version strings are logged. Local model digests and prompt hashes improve reproducibility but cannot make commercial inference fully deterministic.

The next acceptance steps are physician approval of scenario content and paths, independent transcript and evidence audits, resolution of validator and terminology gaps, and evaluation of the deployed authentication and EHR integration boundary. Clinical effectiveness should be studied separately using an approved protocol; it cannot be inferred from software completion or conformance rates.

5 Conclusion

A modular pre-consultation implementation can make questionnaire control, model configuration, provenance, and FHIR write-back inspectable and testable. The present evidence supports a reproducible technical verification workflow, with explicit failures and pending review. It does not establish diagnostic accuracy, clinical safety, or national-scale deployment readiness.

Supplementary information. The repository contains pinned versions, source-level synthetic-data mappings, scenario review packets, model prompts and disclosure,

Table 7 Principal pinned software versions. Model configurations and full artifact digests accompany the experiment manifests.

Component	Pinned version
Synthea	v4.0.0
FHIR	4.0.1
HAPI image	hapiproject/hapi:v8.8.0-1
TW Core IG	1.0.0
HL7 validator	6.10.4
Python	3.12.15
Node	v22.23.2
Ollama	0.32.6
default extraction	gemini-3.6-flash
default summary extraction	gemini-3.5-flash-lite
default summary reasoning	gemini-3.6-flash
swap a	gemini-2.5-flash
swap b	gemma3:12b
patient simulator model	qwen2.5:7b-instruct

raw run manifests, audit samples, and generated tables. Table 7 lists principal software versions; complete digests are in the machine-readable lock file.

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Declarations

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Competing interests. Authors must provide their declarations.

Ethics and consent. These verification runs use synthetic data only. No claim of institutional ethics approval or exemption is made; authors must confirm the applicable institutional determination before submission.

Data and code availability. Scripts, configuration, derived summaries, and asset generators are in the project repository. Large synthetic bundles and raw local runs are regenerable and excluded from source control. A public archival identifier and access statement require author confirmation.

Author contributions. Contributions and final approval require author confirmation.

Use of language models. Language models assisted implementation, synthetic scenario drafting, experiment orchestration, and manuscript preparation. System and simulated-patient configurations, prompt hashes, and run dates are recorded in `experiments/llm_disclosure.json`. Physician scenario review and transcript audits remain pending; no model output is represented as human validation. Authors retain responsibility for source verification and the final manuscript.

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